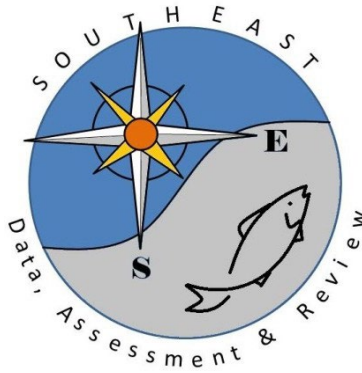




# SEDAR

Southeast Data, Assessment, and Review

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## SEDAR 79

### Southeastern US Mutton Snapper

#### SECTION IV: Research Recommendations

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## 1. DATA WORKSHOP RESEARCH RECOMMENDATIONS

### 1.1 LIFE HISTORY RESEARCH RECOMMENDATIONS

#### 1.1.1 Stock Definition

The LHW recommended expanding genetic sampling to other areas parts of the defined unit stock (e.g., northeastern Florida or on the West Florida Shelf) to either reinforce or challenge current hypothesized boundaries of the Mutton Snapper stock within southeastern U.S. waters. In addition, the presence of Mutton Snapper larvae sampled at stations across the Straits of Florida between the east Florida shelf (e.g., off Biscayne Bay) and the Great Bahama Bank (D'Alessandro et al. 2010) suggests possible connectivity between the two regions. There is no genetic data published from Mutton Snapper in the Bahamas (Carson et al. [2011] comments about it being a less well-documented aggregation area) and investigating this could provide insight into any connection with southeastern Florida as well as any potential source and sink dynamics. While genetic analyses for Mutton Snapper have been conducted from elsewhere in the Caribbean (e.g., Puerto Rico, St. Croix, St. Thomas, Cuba, Belize) and support a homogenous stock, there is no genetic information available from populations observed off the Yucatan peninsula in the southwest Gulf of Mexico.

1.1.2 Morphometrics and Conversion Factors. The LHW recommends additional length measurements in 'maximum' total length be taken across both fishery-dependent and fishery-independent programs to better align with current management regulations and the length units used within the assessment model(s).

#### 1.1.3 Age and Growth

The number of otoliths sampled and available for this assessment had significantly increased from the number available in the previous update assessment and was adequate for use in developing models for growth and tracking strong year classes through time (albeit once they've entered the fishery, about age-3). Yet, the LHW noted a paucity of fishery-independent age data particularly for pre-fishery individuals aged 0 – 2 years and also greater than age ~8 years. Of the ongoing fishery-independent surveys which track young Mutton Snapper throughout the Florida Keys and southeast Florida regions, otolith-derived age information was largely collected from the Indian River Lagoon system.

The LHW, therefore, expressed an interest in the need to increase fishery-independent age sampling of these younger and older parts of the population. Such information could help further our understanding of ontogeny and recruitment throughout the Florida Keys and allow for earlier detection of strong year classes, rather than waiting for them to be sampled from the fishery. Information of older fish also makes possible fishery-independent estimates of  $L_{\infty}$ , which is currently not feasible. The LHW also understands that implicit in this is the probable expansion of fishery-independent surveys targeting these parts of the Mutton Snapper population and may be considered a 'heavy lift'.

#### 1.1.4 Natural Mortality

The field of natural mortality is not yet in a position to establish ‘best practices’ (Maunder et al. 2023) and suggested ‘good practices’ are often trade-offs between reliability and availability of the data. More direct methods of estimating mortality, such as mark-recapture or acoustic telemetry tagging, are generally recommended over empirical methods but are largely unavailable for Mutton Snapper. Therefore, tagging studies for this species are recommended for the purposes of estimating mortality. Effort into acoustic telemetry tagging requires a large enough array of detectors to minimize incomplete detections and the candidacy of Mutton Snapper may need to be evaluated if movement out of the array area is for extended periods (i.e., are the change in numbers due to mortality or migration). But acoustic telemetry will also help alleviate the human reporting issues within conventional tagging.

#### 1.1.5 Release Mortality

Future research is recommended to obtain a more accurate estimate of Mutton Snapper discard mortality, which could include similar work conducted by Forrestal et al. (2017) on the development of physiological parameters or acoustic tagging of Mutton Snapper releases at a series of depths to evaluate post release mortality. In addition, studies that quantify Mutton Snapper immediate and delayed mortality of discards caught from commercial fishing gears are needed to validate the assumption that discard mortalities are comparable among fisheries.

Depredation (the removal of fish from fishing gear by non-target species) of Mutton Snapper is a concern among commercial, charter, and headboat captains primarily in the Gulf of Mexico in recent years (GMFMC Staff 2023 and Workgroup discussions). However, depredation is currently not explicitly incorporated in estimates of discard mortality. Attempts should be made to measure depredation rates from either existing or new surveys and provide recommendations on how to incorporate this information in a stock assessment model.

#### 1.1.6 Reproduction

The LHW emphasized that additional research is needed to better understand Mutton Snapper reproduction in the southeastern U.S., and it is important to note that all reproductive data used here were largely more than 10 years old. A common problem sampling for maturity is that truly immature fish are often smaller than legal size and/or are located in habitat differing from adult habitat. Histological data from the Florida Keys, especially from fishery-independent sources on immature fish, is needed and should be collected throughout the year given the recent best practices developed and conducted here for determining size- and age-at-maturity and spawning seasonality. Furthermore, because reproductive timing in spring and summer-spawning fish is tightly coupled to temperature (Lowerre-Barbieri et al. 2011) spawning seasonality may have changed with climate change. Data are also suggestive of potential migration through SE Florida to Keys spawning grounds and possibly even a second spawning season. Therefore, understanding all these processes will be critical to estimating annual fecundity in this species.

### 1.1.7 Movement and Migration

The movement data presented here for Mutton Snapper is recent and unpublished but is already challenging previous understandings for this species. The LHW, therefore, recommended continual investigation of the movement and migration rates between the Florida Keys, southeast Florida, and southwest Florida (e.g., through increased tagging) as well as to continue examining migration distances and catchment areas of Mutton Snapper traveling to known spawning aggregations. The LHW also recommended further investigation into ontogenetic shifting of juveniles from nearshore areas to reef habitat as this is not well documented within south Florida waters.

## 1.2 COMMERCIAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

### Biological Sampling

- Increased observer and EM coverage in the Gulf and South Atlantic fisheries
- Recommend the observer program investigate the allocation of some observer coverage for focused trips to aid in future SEDARs as a subset of existing strata
  - For Mutton Snapper, this may include allocating addition effort to increase the probability of Mutton encounters in areas of higher population density (SE FL, FL Keys)
- Allocating funding to support research on predator depredation and effect on landings and discards

## 1.3 RECREATIONAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

- Study effect of hurricanes on mutton movement
- Continuation of FES pilot study
- Continued evaluation of appropriate SRFS/FES calibration
- Improve precision of MRIP estimates for the shore mode
- Study rates of depredation on recreational fishing

## 1.4 MEASURES OF POPULATION ABUNDANCE RESEARCH RECOMMENDATIONS

- The IWG recommends analyzing past and future videos from stereo cameras deployed as part of the Southeast Reef Fish Survey (SERFS) for Mutton Snapper lengths.
- Explore methods to incorporate lengths sampled by the Biscayne Bay Creel Survey into the MRIP APAIS.
- Add Mutton Snapper to FWC FIM's standard cull list to collect otoliths from Mutton Snapper sampled by the 183-m seine survey in the Indian River Lagoon.
- Consider incorporating power analyses for other indices similar to the exploratory power analysis of the FIM inshore seine survey.

## 2. ASSESSMENT PROCESS RESEARCH RECOMMENDATIONS

This assessment, as well as many others, would greatly benefit from a better understanding of movement and stock structure, as well as recent reproduction. A better understanding of these processes may shed light on the mechanisms driving the truncated length and age distributions of Mutton Snapper observed in southeast FL. The SEDAR 79 Data Workshop report includes an extensive list of research recommendations from each working group that should be prioritized (Sections 2.11, 3.8, 4.8, 5.6).

Regarding fishery sampling effort, the next assessment would benefit from additional sampling in the FL Keys, increased sampling of commercial vertical line and other gears, additional age and length samples from private and shore recreational modes, and more information on release sizes. The length and age distributions of landed Mutton Snapper in the FL Keys appear to be intermediary between generally smaller/younger fish caught in SE FL and larger/older fish caught in the Dry Tortugas, however the FL Keys are considerably under sampled in most years. For commercial vertical line and other gears, increase the number of measured Mutton Snapper to at least 320 per year. An average of approximately 20 per month from January through August and 40 per month from September through December, especially from vessels fishing in the FL Keys and Dry Tortugas. Additional age and length samples from private and shore recreational modes are needed. Increase length sampling in all regions for the shore mode and begin aging Mutton Snapper landed by the shore mode. For both the private and shore recreational modes, increase length sampling substantially in regions west of the FL Keys and the FL Keys to at least 150 per year and continue sampling the SE FL with the goal of reaching at least 150 per year. Also, the private and shore modes account for nearly all releases but do not contribute release lengths, undoubtedly due to the logistical challenges of sampling releases from these fishing modes.

Lastly, reliable estimates of commercial discards are currently lacking. Estimates of commercial discards from at-sea observers or other sources that are demonstrated to be more reliable than discard logbooks are needed.

### **3. REVIEW PANEL RESEARCH RECOMMENDATIONS**

The RP supports the research recommendations identified by the Data and Assessment stages for the Southeastern US Mutton Snapper assessment processes.

The RP also recommends that this stock is assessed sooner than 10 years given the uncertainties associated with the scale of the population. The dome-shaped selectivity of the recreational fishery on the east coast of Florida is especially concerning. The stock may be overestimated if larger snapper are underrepresented in the recreational catch monitoring programs.

The following recommendations are ordered by priority.

Short-term (within six months)

- Continue efforts to generate MCMC samples to refine the estimation of parameter uncertainty and support stochastic projections.

- Consider statistical methods for making probabilistic statements regarding current status of the stock that accounts for uncertainty in both the reference points and stock size or rates. MCMC methods would provide the necessary joint samples for this computation. Alternatively, it may be possible to use the variance covariance matrix to approximate the probability of being above or below a given reference point.

#### Longer-term

- The model estimated strong dome-shaped selectivity curves for both recreational sectors (east and west), but there remains uncertainty about whether those older/larger fish are less vulnerable to recreational fishing, or whether they are missing from the recreational catch data. It was discussed that Mutton Snapper are often caught as part of an aggregate snapper-grouper trips, and the lack of schooling during most of the year likely precludes high catches as a regular occurrence. Further, some fraction of the offshore recreational vessels use private docks and launches for their trips. Finally, most length-composition information on the recreational fishery comes from monitoring of headboat removals. Thus, samples currently collected in the recreational surveys may not be representative of the fish actually taken in the recreational fishery. We believe improved methods and data collection are needed to better quantify the size/age composition of the recreational sectors.
- A study of the size/age based vertical distribution of Mutton Snapper would be helpful to better understand the availability of fish to fishing gear and the video sampling equipment.
- Consider alternative approaches for longline catch rate standardization, ideally allowing the most recent (2011 to 2023) data to be included.
- Explore the application of spatiotemporal models to the combined Gulf video survey data to monitor distribution shifts and generate an index that accounts for the changes in the survey.
- Though it would be ideal to estimate movement rates and work toward a spatial assessment model, a simpler model that accounts for coarse spatial differences could still be constructed. For instance, a two-box model estimating the portion of each age group inhabiting the West vs East might be feasible. A well-designed conventional tagging program could be a cost-effective and efficient method to help inform this analysis.
- In addition, a high and low- reward floy tagging program can also be used to inform levels of recreational removals by comparing returns from the commercial fishery relative to the recreational fishery.
- If funds permit, data storage tags could be deployed to complement the results from the conventional tagging program.
- Consider close-kin-mark-recapture (CKMR). There is an active fishery with length and age sampling, meaning there is scope for collecting genetic samples, and the population is relatively small, meaning CKMR population estimation may be feasible with relatively few samples. These data also help with defining stock boundaries